

## Inigo Quilez :: articles :: Rendering Worlds With Two Triangles - 2008



Thanks to the [demoscene outreach group](#) guys I and a few other demosceners got to give some seminars at [nvscene](#) in San Jose (California) back in 2008. It was some very interesting days, where we enjoyed a nice party, learned a lot from each other, and met the American demoscene. The hotel was in the very same conference center, so it was really cool :) My lecture/presentation was about my recent work on rendering procedural images based on raymarching, in the GPU, in the context of four kilobyte demos.



Raymarching SDFs, 2008

In this presentation first I introduced the four kilobytes demos that had been produced by the demosceners thus far. This was indeed a good chance to show the world what amazing things demosceners had been doing over the decades, so I talked about all sort of 1k and 4k intros featuring raytracing, raymarching and similar techniques. Then I quickly introduced the concept of raymarching through voxels and procedural iso-surfaces. And then I introduced the concept of distance fields as an acceleration technique for raymarching and rendering. This had been a dismissed if not forgotten field of Computer Graphics, and I like to think I somehow brought it to people's attention by showing the making of my Slisesix SDF demo (picture above). For extra kicks, this rendering demo was only 3 kilobytes in size, including the models, textures, lighting, materials, and renderer itself.

During my talk I also introduced some of the most basic operations in procedural SDFs, such as [Domain](#)

**Repetition**, Deformations and **Smooth Blending**. I also explained how to efficiently compute Ambient Occlusion and **Soft Shadows**. I think it was a rather complete and fun presentation, specially because the slides where actually live and interactive. Unfortunately I no longer have the code for the actual slideshow, so instead I'm leaving a backed PDF version of it here for you to download:

**Rendering Worlds With Two Triangles PDF**

inigo quilez - learning computer graphics since 1994